

Problem-Solution Fit

Date	29 June 2026
TeamID	xxxxxx
ProjectName	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-SolutionFit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] 1) Farmers 2) Agricultural Land Owners 3) Agricultural Cooperatives 4) Agricultural Extension Officers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] 1) Limited technical knowledge 2) Budget constraints 3) Lack of access to experts 4) Changing weather conditions 5) Difficulty understanding soil reports	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional Farming Experience 1) Pros: Simple and familiar 2) Cons: Not data-driven, less accurate Agricultural Experts 1) Pros: Professional advice 2) Cons: Limited availability and expensive Government Advisory Services 1) Pros: Reliable information 2) Cons: Generic recommendations
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2. PROBLEMS / PAINS

+ ITS FREQUENCY [PR]

- 1) Difficulty selecting the right crop for soil conditions (High)
- 2) Low crop yield due to poor decision-making (High)
- 3) Excessive fertilizer usage increases cost (Medium)
- 4) Unpredictable weather affects farming (High)

9. PROBLEMROOT/

CAUSE [RC]

- 1) Lack of data-driven crop selection
- 2) Insufficient knowledge of soil nutrients
- 3) Climate variability
- 4) Dependence on traditional farming methods

7. BEHAVIOR

+ ITS INTENSITY [BE]

- 1) Frequently checks weather forecasts
- 2) Relies on previous farming experience
- 3) Consults local farmers before planting
- 4) Intensity: High (every farming season)

3. TRIGGERSTOACT

[TR]

- 1) Poor harvest in previous seasons
- 2) Soil test report availability
- 3) Rising fertilizer and cultivation costs
- 4) Need to maximize profit and productivity

10. YOURSOLUTION

[SL]

OptiCrop is an AI-powered crop recommendation system that analyzes soil nutrients (N, P, K, pH) and environmental factors (temperature, humidity, rainfall) using a Random Forest machine learning model to recommend the most suitable crop. This helps farmers improve crop yield, reduce cultivation costs, and make informed farming decisions.

8. CHANNELSOF

BEHAVIOR [CH]

ONLINE

- 1) Agriculture websites
- 2) Mobile applications
- 3) YouTube
- 4) WhatsApp farmer groups

OFFLINE

- 1) Local agriculture offices
- 2) Soil testing laboratories
- 3) Farmer meetings
- 4) Agricultural extension centers

4. EMOTIONS

BEFORE / AFTER [EM]

Before:

- 1) Confused
- 2) Worried
- 3) Uncertain

After:

- 1) Confident
- 2) Satisfied
- 3) Hopeful